UnextendedEC A2 | NIZ EC + ANSI 6.25u + nicefnano v2 | ENGINEERING PROTOTYPE

Native nicefnano USB-C is the external port. J2 carries no USB. VBUS or high-impedance SENSE.

U2/U3: only one INH low at a time. U4 gain 101. DRAIN goes through R32 to SENSE. Q1 gates U4 only.

SW1/SW2 are 2.54mm solder-hole harnesses. No external 5V connection to U1 BATTERY+.

J2
GND
GND
VCC
GND
ROW_DRV0
ROW_DRV1
ROW_DRV2
ROW_DRV3
ROW_DRV4
GND
MUX_A
MUX_B
MUX_C
MUX1_INH
MUX2_INH
GND
DRAIN
AFE_ON_N
GND
ADC
GND
BAT_SW
RESET
GND
MUX_C
MUX2_INH
MUX1_INH
GND
DRAIN
AFE_ON_N
GND
ADC
GND
BAT_SW
RESET
GND
FH12-24S-0.5SH

U2
COL9
COL10
SENSE1
COL7
COL11
MUX2_INH
GND
VEE
GND
VCC
X6
COM
X7
X5
INH
VEE
GND
C
B
A
X3
X0
X1
X2
VCC
MUX_C
MUX_B
MUX_A
COL5
COL8
COL6
COL4
VCC
CD74HC4051M96

U3
COL12
COL14
SENSE2
GND
COL13
MUX1_INH
GND
VEE
GND
VCC
X6
COM
X7
X5
INH
VEE
GND
C
B
A
X3
X0
X1
X2
VCC
MUX_C
MUX_B
MUX_A
COL15
COL9
COL10
COL8
VCC
CD74HC4051M96

U4
FB
SENSE
GND
NC
IN-
IN+
V-
NC
OUT
V+
NC
AMP_OUT
AFE_3V3
OPA3501A

Q1
AFE_GATE
VCC
G
S
D
AFE_3V3
AO3401A

J1
BAT_RAW
BAT_RAW_GND
GND
JST_S2B-PH-K

SW1
BAT_RAW
BAT_RAW_SW
BAT_SW
Battery_switch_pigtail

SW2
RESET
RESET GND
GND
Reset_NO

R1 ROW_DRV0 ROW0 100
R2 ROW_DRV1 ROW1 100
R3 ROW_DRV2 ROW2 100
R4 ROW_DRV3 ROW3 100
R5 ROW_DRV4 ROW4 100
R30 AMP_OUT FB 100k
R31 FB GND 1k
R32 DRAIN SENSE 1k
R33 MUX1_INHC VCC 100k
R34 MUX1_INHC VCC 100k
R35 AFE_GATE VCC 100k
R36 AFE_ON_N AFE_ON_BAT 1k
R37 AMP_OUT ADC 47

R10 COL0 GND 100k
R11 COL1 GND 100k
R12 COL2 GND 100k
R13 COL3 GND 100k
R14 COL4 GND 100k
R15 COL5 GND 100k
R16 COL6 GND 100k
R17 COL7 GND 100k
R18 COL8 GND 100k
R19 COL9 GND 100k
R20 COL10 GND 100k
R21 COL11 GND 100k
R22 COL12 GND 100k
R23 COL13 GND 100k
R24 COL14 GND 100k

C1 VCC GND 100n
C2 VCC GND 100n
C3 AFE_3V3 GND 100n
C4 SENSE GND 220p_C0G
C5 VCC GND 1u

EC electrodes: logical row = physical row; column = floor(x_u + width_u/2). Including empty column blns.

EC1 ROW0 COL0 DRIVE GUARD SENSE GND Esc
EC2 ROW0 COL1 DRIVE GUARD SENSE GND 1
EC3 ROW0 COL2 DRIVE GUARD SENSE GND 2
EC4 ROW0 COL3 DRIVE GUARD SENSE GND 3
EC5 ROW0 COL4 DRIVE GUARD SENSE GND 4
EC6 ROW0 COL5 DRIVE GUARD SENSE GND 5
EC7 ROW0 COL6 DRIVE GUARD SENSE GND 6
EC8 ROW0 COL7 DRIVE GUARD SENSE GND 7
EC9 ROW0 COL8 DRIVE GUARD SENSE GND 8
EC10 ROW0 COL9 DRIVE GUARD SENSE GND 9
EC11 ROW0 COL10 DRIVE GUARD SENSE GND 0
EC12 ROW0 COL11 DRIVE GUARD SENSE GND -
EC13 ROW0 COL12 DRIVE GUARD SENSE GND =
EC14 ROW0 COL13 DRIVE GUARD SENSE GND Backspace
EC15 ROW0 COL14 DRIVE GUARD SENSE GND Tab
EC16 ROW1 COL0 DRIVE GUARD SENSE GND Q
EC17 ROW1 COL1 DRIVE GUARD SENSE GND W
EC18 ROW1 COL2 DRIVE GUARD SENSE GND E
EC19 ROW1 COL3 DRIVE GUARD SENSE GND R
EC20 ROW1 COL4 DRIVE GUARD SENSE GND T
EC21 ROW1 COL5 DRIVE GUARD SENSE GND Y
EC22 ROW1 COL6 DRIVE GUARD SENSE GND U
EC23 ROW1 COL7 DRIVE GUARD SENSE GND I
EC24 ROW1 COL8 DRIVE GUARD SENSE GND O
EC25 ROW1 COL9 DRIVE GUARD SENSE GND P
EC26 ROW1 COL10 DRIVE GUARD SENSE GND [
EC27 ROW1 COL11 DRIVE GUARD SENSE GND]
EC28 ROW1 COL12 DRIVE GUARD SENSE GND \
EC29 ROW1 COL13 DRIVE GUARD SENSE GND Caps
EC30 ROW1 COL14 DRIVE GUARD SENSE GND A
EC31 ROW2 COL0 DRIVE GUARD SENSE GND S
EC32 ROW2 COL1 DRIVE GUARD SENSE GND D
EC33 ROW2 COL2 DRIVE GUARD SENSE GND F
EC34 ROW2 COL3 DRIVE GUARD SENSE GND G
EC35 ROW2 COL4 DRIVE GUARD SENSE GND H
EC36 ROW2 COL5 DRIVE GUARD SENSE GND J
EC37 ROW2 COL6 DRIVE GUARD SENSE GND K
EC38 ROW2 COL7 DRIVE GUARD SENSE GND L
EC39 ROW2 COL8 DRIVE GUARD SENSE GND ;
EC40 ROW2 COL9 DRIVE GUARD SENSE GND '
EC41 ROW2 COL10 DRIVE GUARD SENSE GND Enter
EC42 ROW2 COL11 DRIVE GUARD SENSE GND Shift
EC43 ROW2 COL12 DRIVE GUARD SENSE GND Z
EC44 ROW2 COL13 DRIVE GUARD SENSE GND X
EC45 ROW2 COL14 DRIVE GUARD SENSE GND C
EC46 ROW3 COL0 DRIVE GUARD SENSE GND V
EC47 ROW3 COL1 DRIVE GUARD SENSE GND B
EC48 ROW3 COL2 DRIVE GUARD SENSE GND N
EC49 ROW3 COL3 DRIVE GUARD SENSE GND M
EC50 ROW3 COL4 DRIVE GUARD SENSE GND ,
EC51 ROW3 COL5 DRIVE GUARD SENSE GND .
EC52 ROW3 COL6 DRIVE GUARD SENSE GND /
EC53 ROW3 COL7 DRIVE GUARD SENSE GND RShift
EC54 ROW3 COL8 DRIVE GUARD SENSE GND LCTRL
EC55 ROW3 COL9 DRIVE GUARD SENSE GND LGUI
EC56 ROW3 COL10 DRIVE GUARD SENSE GND LAlt
EC57 ROW3 COL11 DRIVE GUARD SENSE GND Space
EC58 ROW3 COL12 DRIVE GUARD SENSE GND RAlt
EC59 ROW3 COL13 DRIVE GUARD SENSE GND Fn
EC60 ROW3 COL14 DRIVE GUARD SENSE GND Menu
EC61 ROW4 COL14 DRIVE GUARD SENSE GND RCTRL

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